

## 2017-Ph-H1-Q16

**Section:** Particles and Waves

**Topic:** Nuclear Reactions

A fusion reaction releases energy. Given mass loss, calculate energy released.

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 **Use Einstein's equation:**

$$E = mc^2$$

Where:

- $m = 1.11 \times 10^{-28} \text{ kg}$
- $c = 3.0 \times 10^8 \text{ m/s}$

$$E = 1.11 \times 10^{-28} \times (3.0 \times 10^8)^2 = 1.11 \times 10^{-28} \times 9.0 \times 10^{16} = 1.0 \times 10^{-11} \text{ J}$$

 **Final Answer: B —  $1.0 \times 10^{-11} \text{ J}$**

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 **Revision Tips:**

- Always convert mass to **kg**
  - Fusion and fission both release energy due to **mass defect**
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